

Integers, Place Value and Arithmetic

Module 2 · Homework

Goal: Answer integer and place-value questions fast, with zero errors.

Time target: 25 minutes for the whole set (about 2 minutes per question, 6 for the challenge). Work without a calculator. Circle one answer for each question. Then check with the answer strip and fill in the error log for every wrong answer.

1. *Retrieval · M1 — contest scoring (5/6/8).* In a 25-question contest, questions 1–10 are worth 5 points each, 11–20 are worth 6 points each and 21–25 are worth 8 points each. Priya gets questions 1–10 right and five of the questions 11–20 right. All her other answers are wrong. What is her score?
(A) 75 (B) 80 (C) 90 (D) 110 (E) 120

2. *Retrieval · M1 — try a small case.* What is the value of $1 + 3 + 5 + 7 + \cdots + 39$?
(A) 210 (B) 380 (C) 390 (D) 400 (E) 420

3. *Retrieval · M1 — work backward.* Tomas thinks of a number. He multiplies it by 4 and then subtracts 9. His result is 55. What number did he think of?
(A) 11.5 (B) 16 (C) 46 (D) 64 (E) 229

4. *Current topic · M2.* Which of the following numbers is the smallest?
(A) 0.7 (B) 0.707 (C) 0.077 (D) 0.0707 (E) 0.07

5. *Current topic · M2.* A rope 4.8 m long is cut completely into pieces 30 cm long. An identical rope is cut completely into pieces 40 cm long. How many more 30 cm pieces are there than 40 cm pieces?
(A) 4 (B) 10 (C) 12 (D) 16 (E) 28

6. *Current topic · M2.* A parking lot charges \$1.75 for every 15 minutes or part of 15 minutes. A car parks from 9:52 a.m. until 11:20 a.m. How much does it pay?
(A) \$8.75 (B) \$10.27 (C) \$10.50 (D) \$12.25 (E) \$15.75
7. *Current topic · M2.* At midnight the temperature was -6°C . By 4 a.m. it had fallen 9 degrees. By noon it had risen 23 degrees from its 4 a.m. value. What was the temperature at noon?
(A) -20°C (B) -8°C (C) 8°C (D) 20°C (E) 38°C
8. *Interleaved · M1 work backward + M2 time.* Maya's bus comes at 8:17 a.m. She needs 18 minutes to get ready, 25 minutes for breakfast and 12 minutes to walk to the stop, and she wants to be at the stop 4 minutes early. What is the latest time she can get up?
(A) 7:08 a.m. (B) 7:18 a.m. (C) 7:22 a.m. (D) 7:28 a.m. (E) 7:58 a.m.
9. *Interleaved · M1 eliminate options + M2 middle value.* Which of the following could be the sum of five consecutive whole numbers?
(A) 52 (B) 63 (C) 74 (D) 81 (E) 95

★ **Challenge**

10. *Challenge · M2 — Part C.* The number N is written with fifty 9s: $N = 999\dots 9$. What is the sum of the digits of $7 \times N$?
(A) 350 (B) 441 (C) 450 (D) 451 (E) 459

Self-check answer strip

Final answers only. Fold this page under before you start; unfold it when you have finished.

Q	1	2	3	4	5	6	7	8	9	10
Answer	B	D	B	E	A	C	C	B	E	C

Error log

One row for every wrong answer. Codes: **C** Concept · **R** Representation · **S** Strategy · **A** Arithmetic · **V** Visual/spatial · **Q** Question-reading · **T** Time · **K** Knew-but-didn't-recall

Q	My answer	Correct	Code	What I will do differently